# Supervised Learning (Maths Part) Past Paper Questions

1. $\beta_1$ calculation / Simple Linear Regression

2. Multiple Regression prediction

3. Decision Tree concepts

4. Gini Index calculations

5. Decision Tree classification

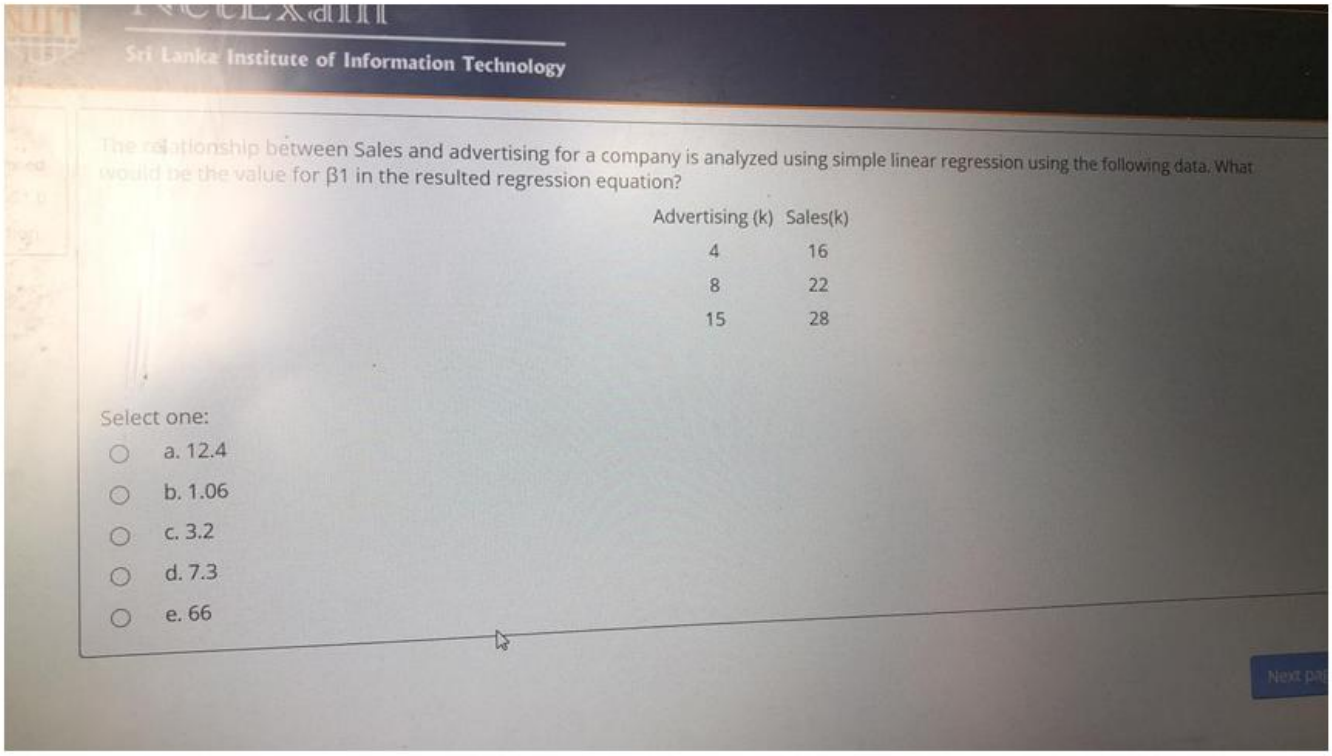
6. Decision Tree application

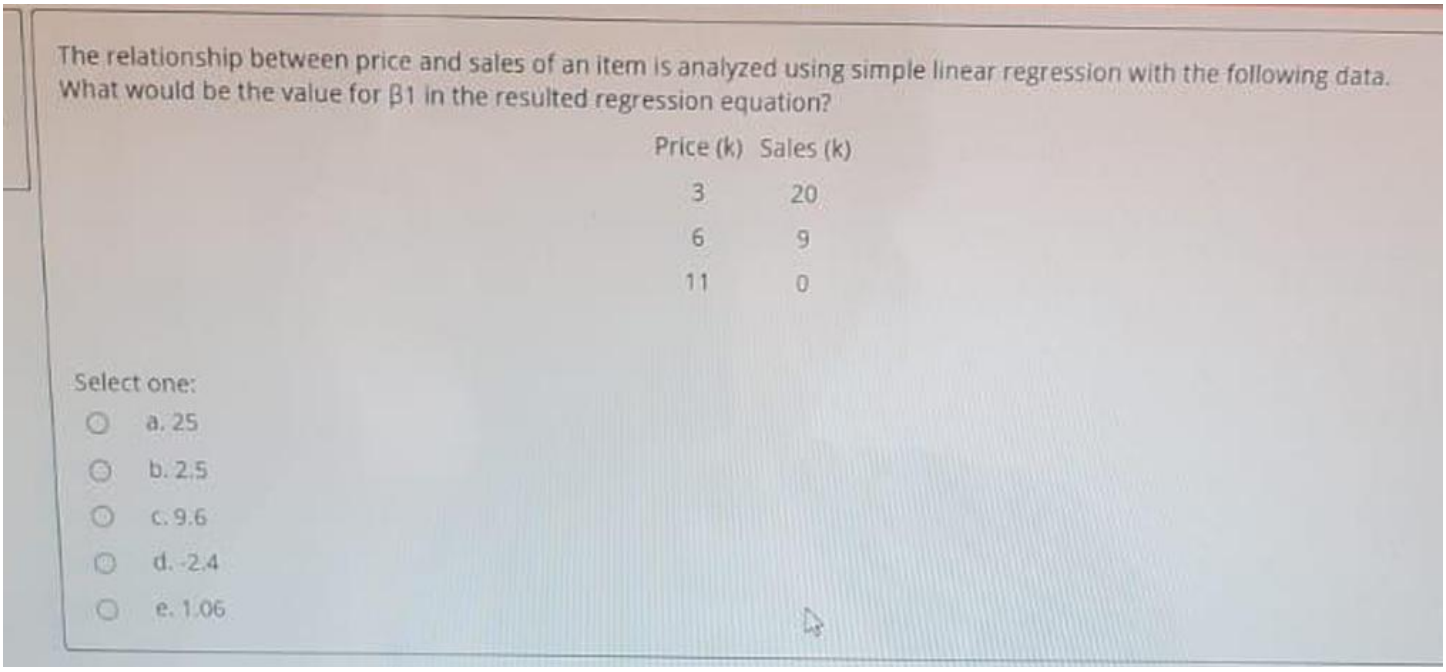
7. $R^2$ / Coefficient of Determination

8. Regression coefficient estimation

9. Linear Regression theory/conditions

# $\beta_1$ calculation / Simple Linear Regression



The relationship between Sales and advertising for a company is analyzed using simple linear regression with th following data. What would be the value for β0 in the resulted regression equation?

| Advertising (k) | Sales(k) |
|---|---|
| 4 | 16 |
| 8 | 22 |
| 15 | 28 |

Select one:

- a. 12.4
- b. 66
- c. 7.3
- d. 1.06
- e. 3.2

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|---|---|
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| 15 | 28 |

Select one:

- a. 12.4
- b. 66
- c. 1.06
- d. 3.2
- e. 7.3

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The relationship between house size and house price is analyzed using simple linear regression using the following data. What would be the value for β1 in the resulted regression equation?

| Size (100) | Price (k) |
|---|---|
| 5 | 14 |
| 13 | 24 |
| 20 | 30 |

Select one:

- a. 1.02
- b. 1.07
- c. 9.18
- d. 9.10
- e. 12.6

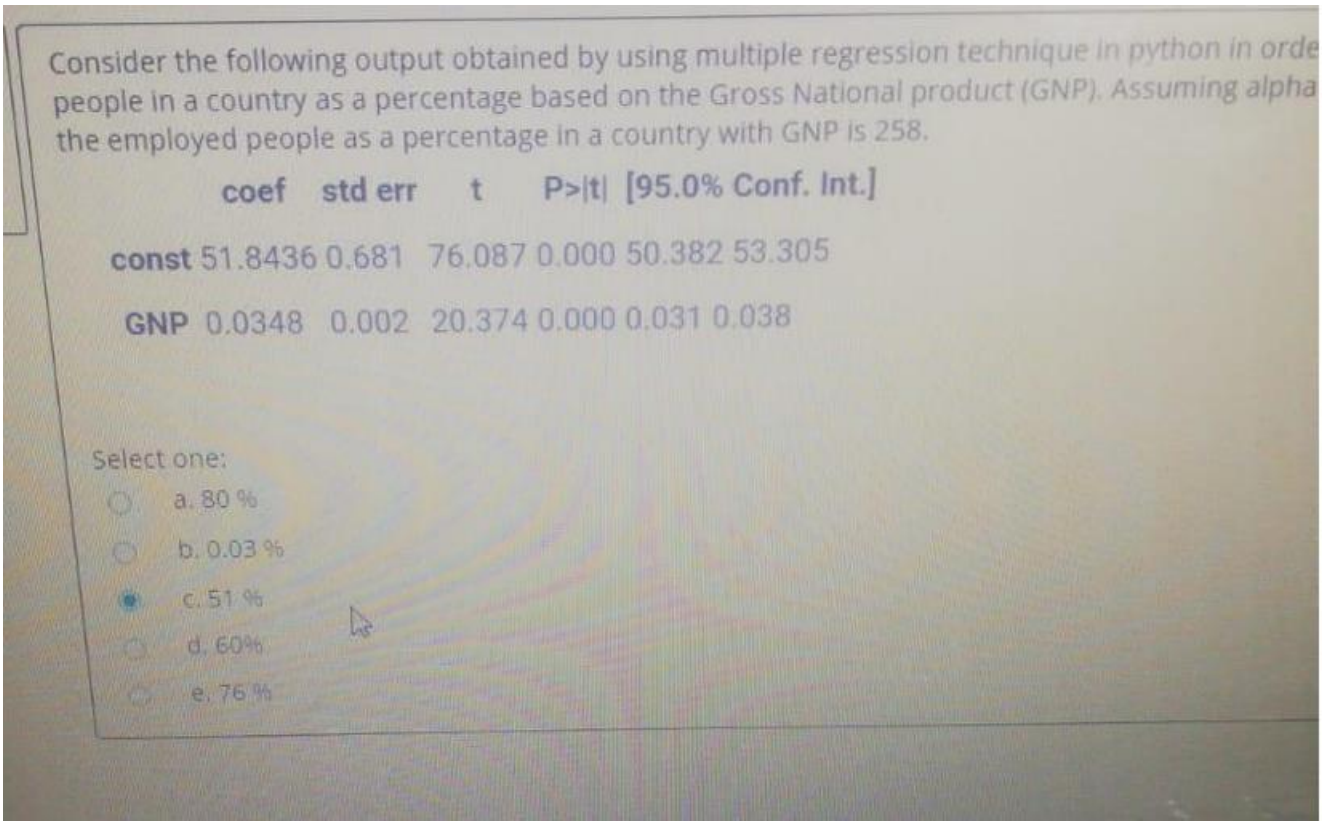
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- d. 12.4
- e. 66

# Multiple Regression prediction

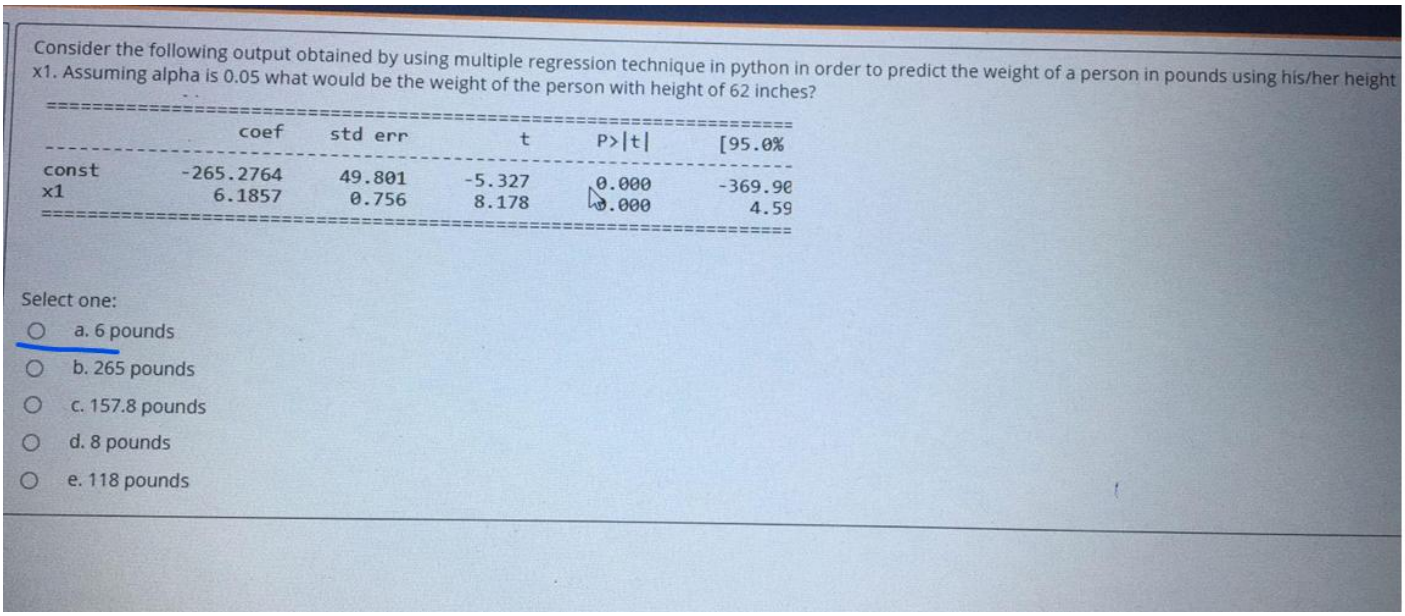


Consider the following output obtained by using multiple regression technique in python in orde
people in a country as a percentage based on the Gross National product (GNP). Assuming alpha
the employed people as a percentage in a country with GNP is 258.

| | coef | std err | t | P>\|t\| | [95.0% Conf. Int.] | |
|---|---|---|---|---|---|---|
| const | 51.8436 | 0.681 | 76.087 | 0.000 | 50.382 | 53.305 |
| GNP | 0.0348 | 0.002 | 20.374 | 0.000 | 0.031 | 0.038 |

Select one:

- a. 80 %
- b. 0.03 %
- c. 51 %
- d. 60%
- e. 76 %



Consider the following output obtained by using multiple regression technique in python in order to predict the weight of a person in pounds using his/her height x1. Assuming alpha is 0.05 what would be the weight of the person with height of 62 inches?

| | coef | std err | t | P>\|t\| | [95.0% |
|---|---|---|---|---|---|
| const | -265.2764 | 49.801 | -5.327 | 0.000 | -369.98 |
| x1 | 6.1857 | 0.756 | 8.178 | 0.000 | 4.59 |

Select one:

- a. 6 pounds
- b. 265 pounds
- c. 157.8 pounds
- d. 8 pounds
- e. 118 pounds

Consider the following output obtained by using multiple regression technique in python in order to predict the value of the property (MDEV) in Millions by using average number of rooms (RM) and status of the population (LSTAT)

| | coef | std err | t | P>\|t\| | [95.0% Conf. Int.] |
|---|---|---|---|---|---|
| RM | 4.9069 | 0.070 | 69.906 | 0.000 | 4.769 5.045 |
| LSTAT | -0.6557 | 0.031 | -21.458 | 0.000 | -0.716 -0.596 |

What would be the predicted value of the property if average number of rooms in a house is 3 status of the population is 9,

Select one:

- a. 69.9 M
- b. 8.2 M
- c. 3.2 M
- d. 4.9 M
- e. 8.8 M

Consider the following output obtained by using multiple regression technique in python in order to predict the weight of a person in pounds using his/her height in inches denoted by x1. Assuming alpha is 0.05 what would be the weight of the person with height of 62 inches?

| | coef | std err | t | P>\|t\| | [95.0% |
|---|---|---|---|---|---|
| const | -265.2764 | 49.801 | -5.327 | 0.000 | -369.90 |
| x1 | 6.1857 | 0.756 | 8.178 | 0.000 | 4.59 |

Select one:

- a. 265 pounds
- b. 157.8 pounds
- c. 8 pounds
- d. 6 pounds
- e. 118 pounds

Consider the following output obtained by using multiple regression technique in python in order to predict the value of the property (MDEV) in Millions by using average number of rooms (RM) and status of the population (LSTAT)

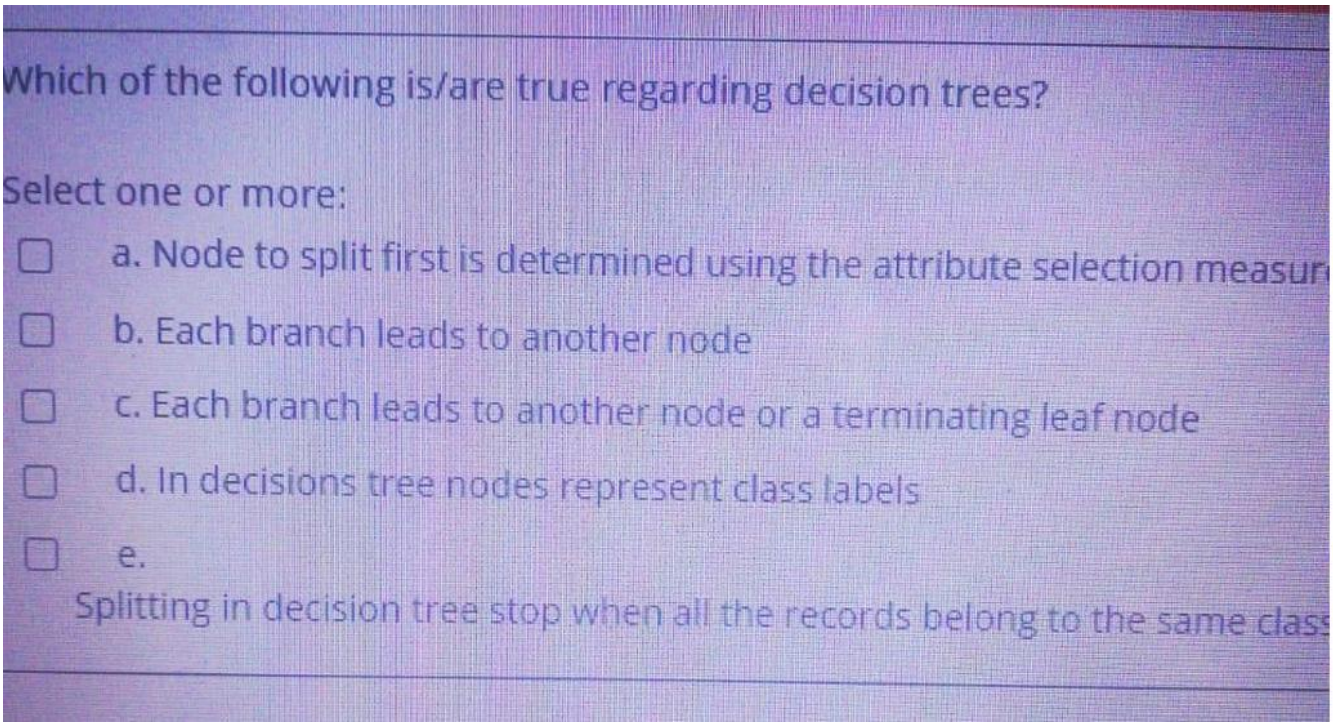
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What would be the predicted value of the property if average number of rooms in a house is 3 status of the population is 9.

Select one:

- a. 4.9 M
- b. 3.2 M
- c. 69.9 M
- d. 8.8 M
- e. 8.2 M

# Decision Tree concepts



Which of the following is/are true regarding decision trees?

Select one or more:

- [ ] a. Node to split first is determined using the attribute selection measur
- [ ] b. Each branch leads to another node
- [ ] c. Each branch leads to another node or a terminating leaf node
- [ ] d. In decisions tree nodes represent class labels
- [ ] e. Splitting in decision tree stop when all the records belong to the same class

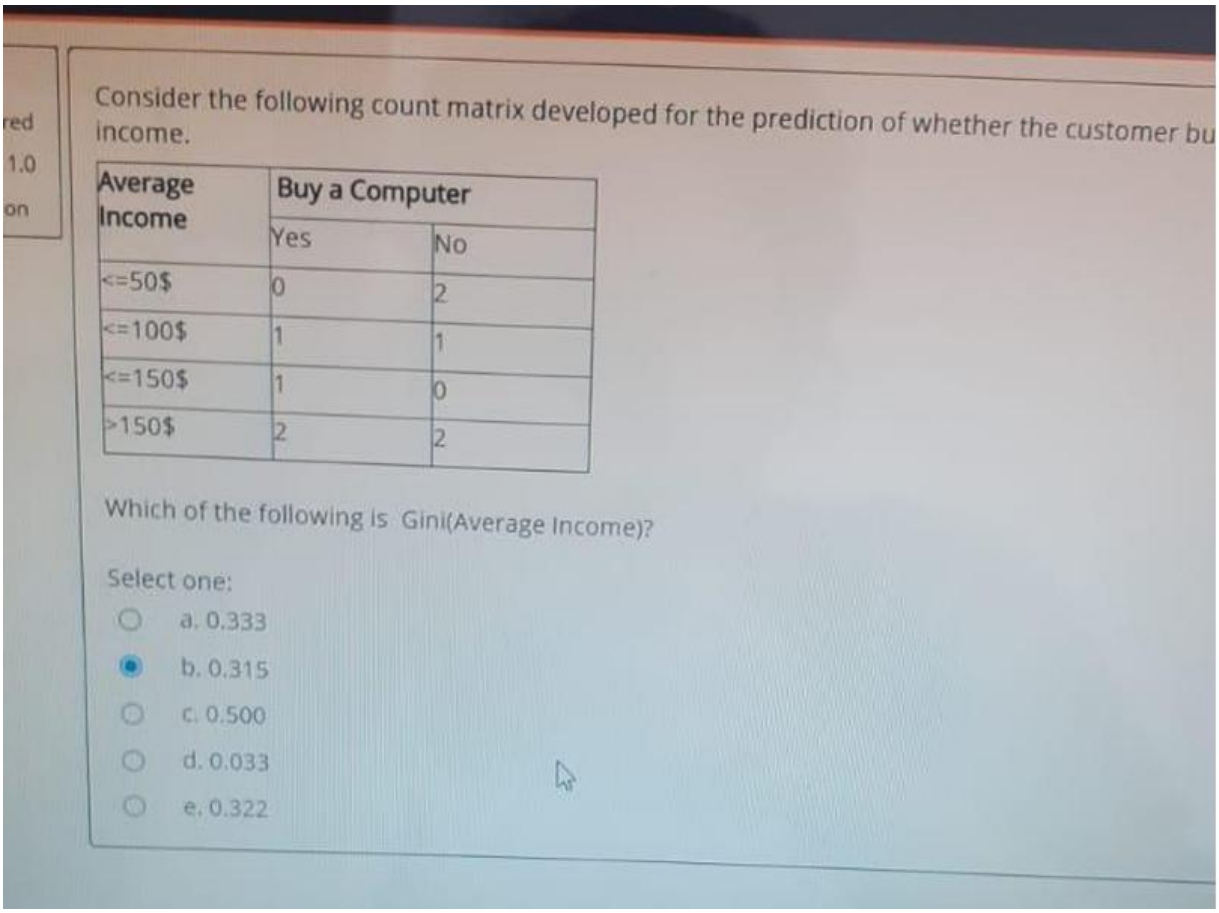
# Gini Index calculations

| Age | Income | Student | Buys_computer |
|---|---|---|---|
| Young | High | No | No |
| Young | High | No | No |
| Middle | High | No | Yes |
| Senior | Medium | No | Yes |
| Senior | Low | Yes | Yes |
| Senior | Low | Yes | No |
| Middle | Low | Yes | Yes |
| Young | Medium | No | No |
| Young | Low | Yes | Yes |
| Senior | Medium | Yes | Yes |
| Young | Medium | Yes | Yes |
| Middle | Medium | No | Yes |
| Middle | High | Yes | Yes |
| Senior | Medium | No | No |

Which of the following is Gini(Age=Middle)?

Select one:

- a. 0.443
- b. 0.245
- c. 0.0
- d. 0.48
- e. 0.375



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1.0
on

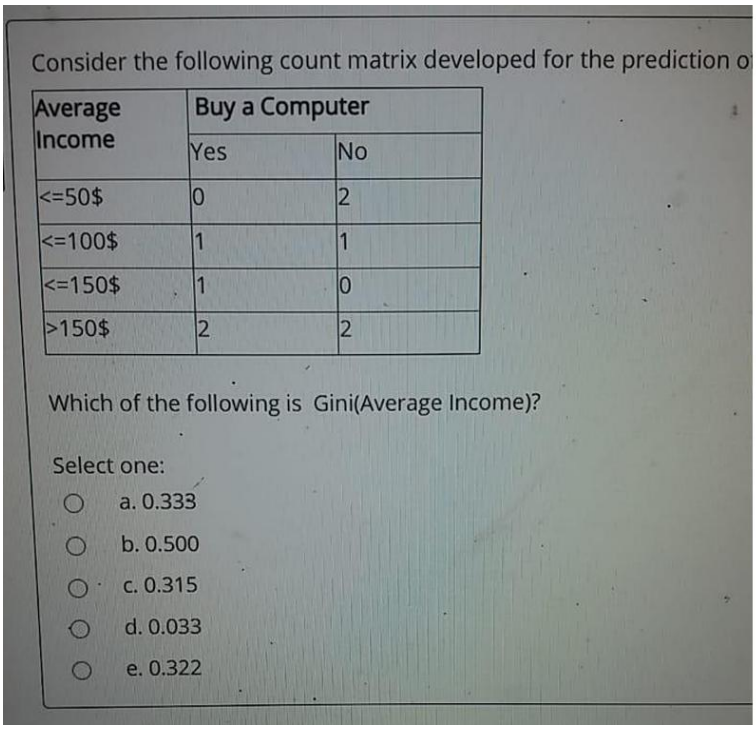
Consider the following count matrix developed for the prediction of whether the customer bu
income.

| Average Income | Buy a Computer | |
|---|---|---|
| | Yes | No |
| <=50$ | 0 | 2 |
| <=100$ | 1 | 1 |
| <=150$ | 1 | 0 |
| >150$ | 2 | 2 |

Which of the following is Gini(Average Income)?

Select one:

- a. 0.333
- b. 0.315
- c. 0.500
- d. 0.033
- e. 0.322



Consider the following count matrix developed for the prediction o

| Average Income | Buy a Computer | |
|---|---|---|
| | Yes | No |
| <=50$ | 0 | 2 |
| <=100$ | 1 | 1 |
| <=150$ | 1 | 0 |
| >150$ | 2 | 2 |

Which of the following is Gini(Average Income)?

Select one:

- a. 0.333
- b. 0.500
- c. 0.315
- d. 0.033
- e. 0.322

Consider the following count matrix developed for the prediction of rain using humidity of the day.

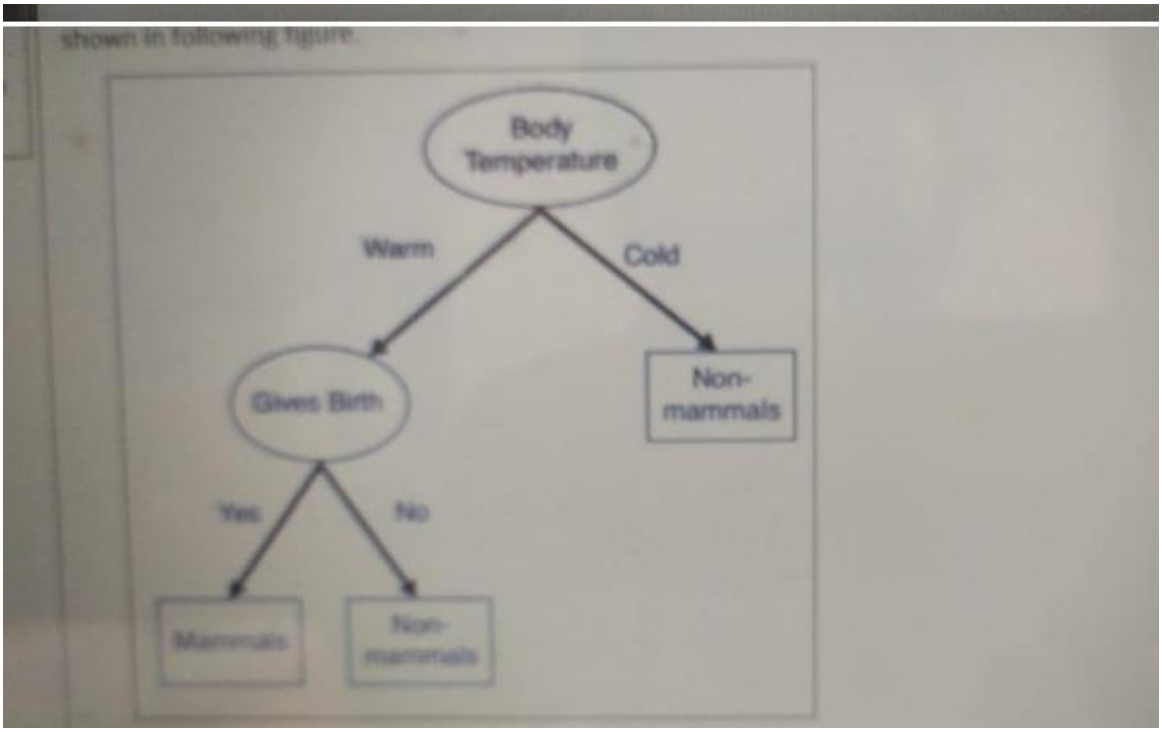
| Humidity | Rain | |
|---|---|---|
| | Yes | No |
| High | 1 | 1 |
| Normal | 2 | 1 |

Which of the following is Gini(Humidity = Normal)?

Select one:

- a. 0.500
- b. 0.444
- c. 0.544
- d. 0.464
- e. 0.400

# Decision Tree classification



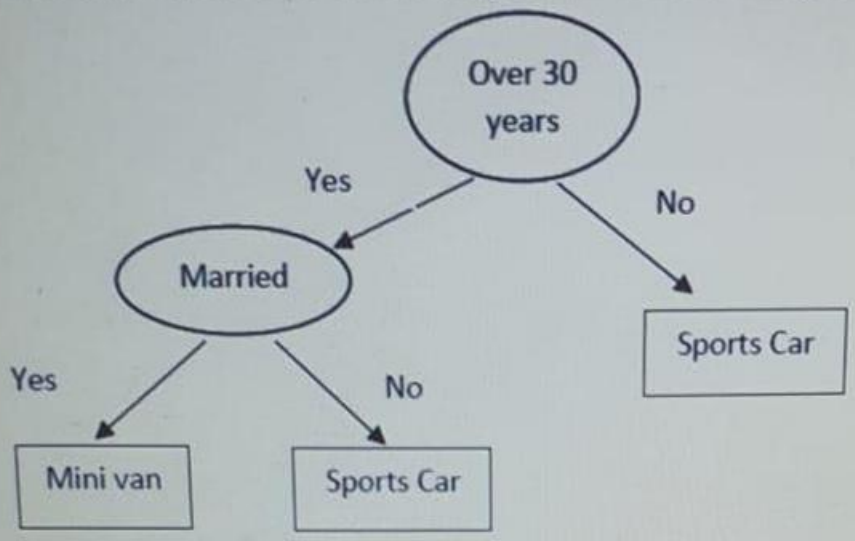

Select the correct answer which contains the class of the unlabeled data shown in following table for the animals in orde
<gibbon, flamingo, lion and cockroach>.

| Name | Body Temperature | Gives Birth | Class |
|---|---|---|---|
| gibbon | warm | yes | ? |
| flamingo | warm | no | ? |
| lion | warm | yes | ? |
| cockroach | cold | no | ? |

Select one:

- a. Non - Mammals, Non-Mammals, Mammals, Non-mammal
- b. Mammals, Non-mammals, Non - Mammals, Non-mammals
- c. Non - Mammals, Mammals, Mammals, Non-mammals
- d. Mammals, Non-mammals, Mammals, Non-mammals
- e. Mammals, Mammals, Mammals, Non-mammals

A decision tree is constructed to predict the type of vehicles a person might buy as follows.



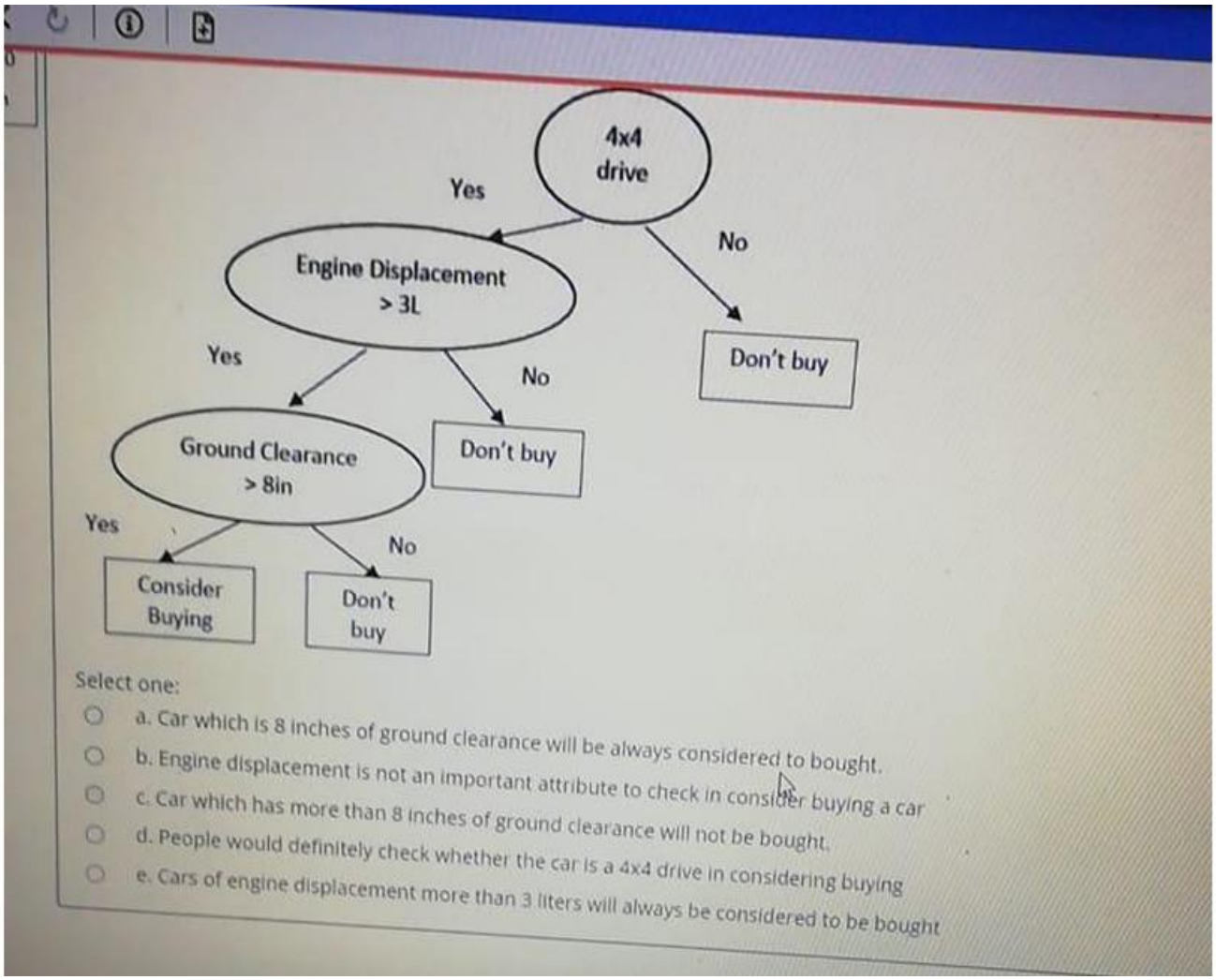

Consider the profiles of following people.

| Name | Age | Marital Status |
|---|---|---|
| Mark | >30 | yes |
| Paul | <=30 | no |
| William | >30 | no |
| Susan | >30 | yes |
| Ronald | <=30 | no |

Which people are likely to buy a sports car?

Select one:

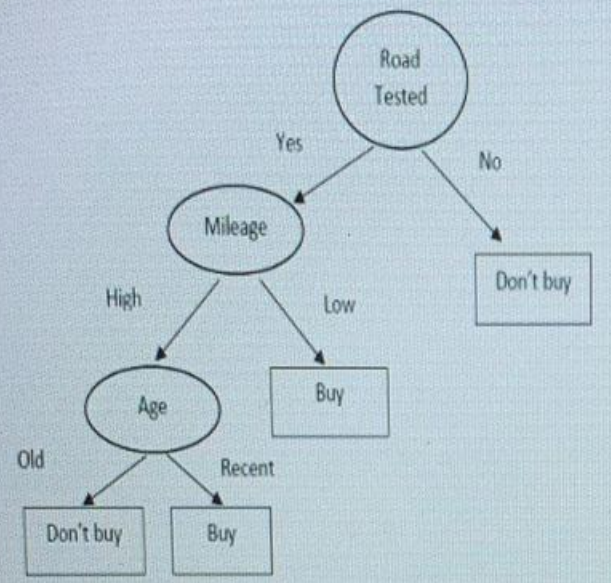
- a. William, Ronald, Mark
- b. Paul, William, Susan
- c. Paul, William, Ronald
- d. Mark, Paul, Ronald
- e. William, Susan, Ronald




Select one:

- a. Car which is 8 inches of ground clearance will be always considered to bought.
- b. Engine displacement is not an important attribute to check in consider buying a car
- c. Car which has more than 8 inches of ground clearance will not be bought.
- d. People would definitely check whether the car is a 4x4 drive in considering buying
- e. Cars of engine displacement more than 3 liters will always be considered to be bought

A decision tree is developed to predict whether the car can be sold (buy or don't buy) by analyzing several attributes such as road tested, age and mileage.

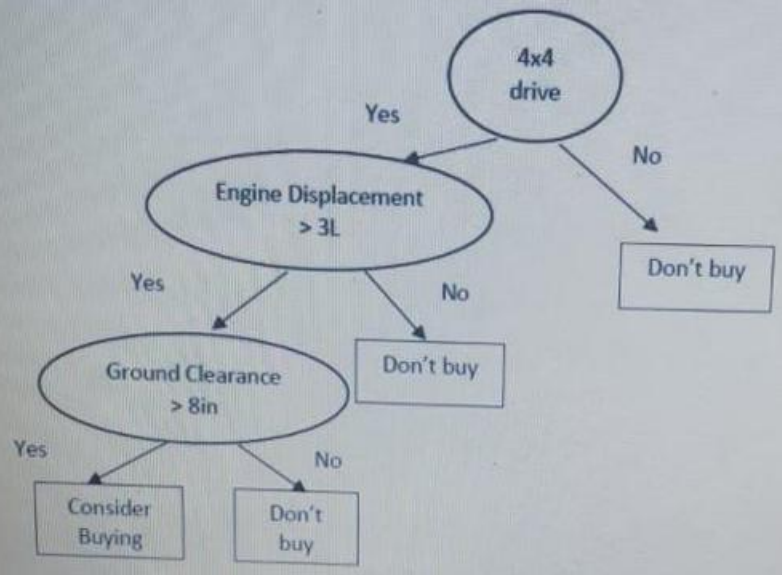



Which of the following statements are correct?

Select one or more:

- [ ] a. When the mileage is high in old cars, people will not buy.
- [ ] b. People would always like to buy a car which has a low mileage
- [ ] c. People would not always consider the age of the car in buying.
- [ ] d. People would not buy old cars.
- [ ] e. People might buy road tested cars.

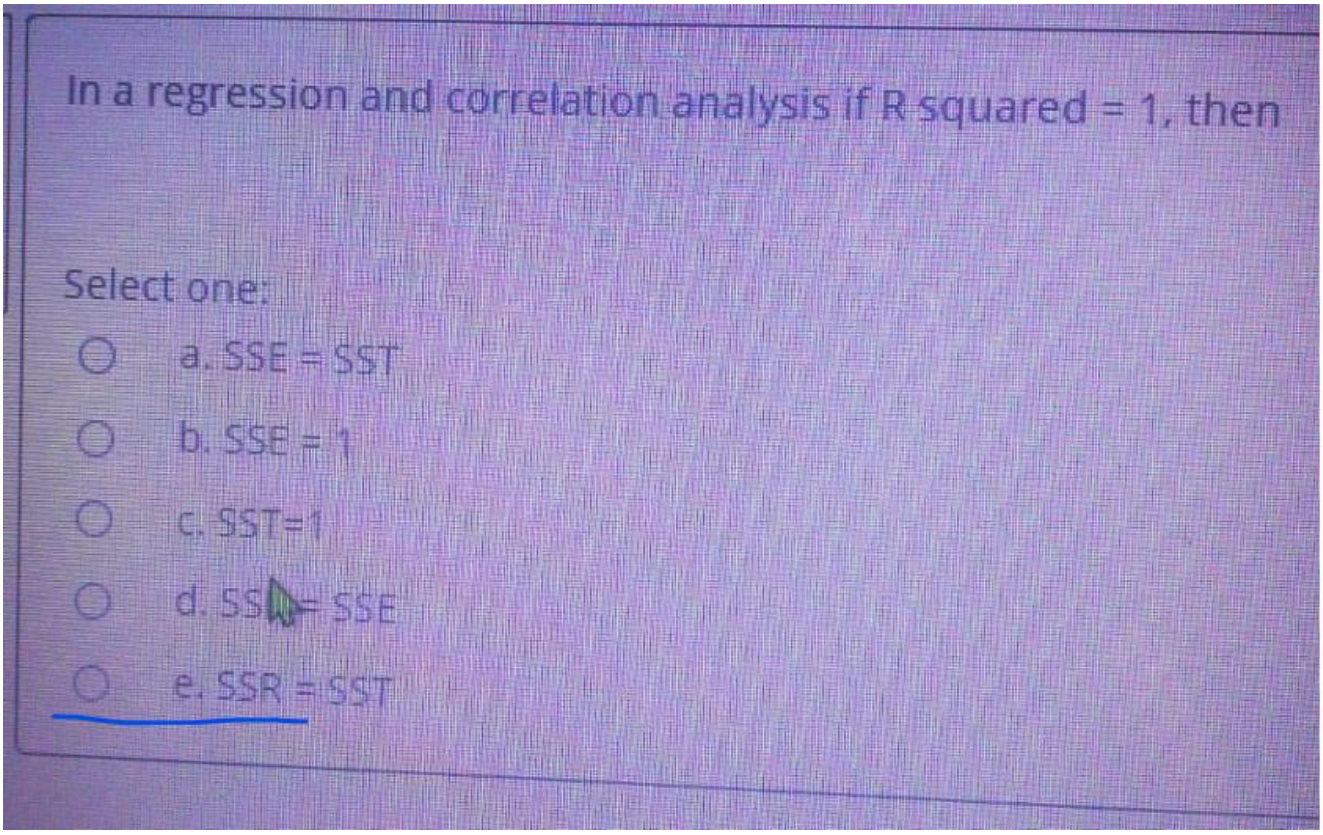
A decision tree is developed as follows to predict whether a car will be bought (consider buying or don't bu
the type of drive, engine displacement and ground clearance.



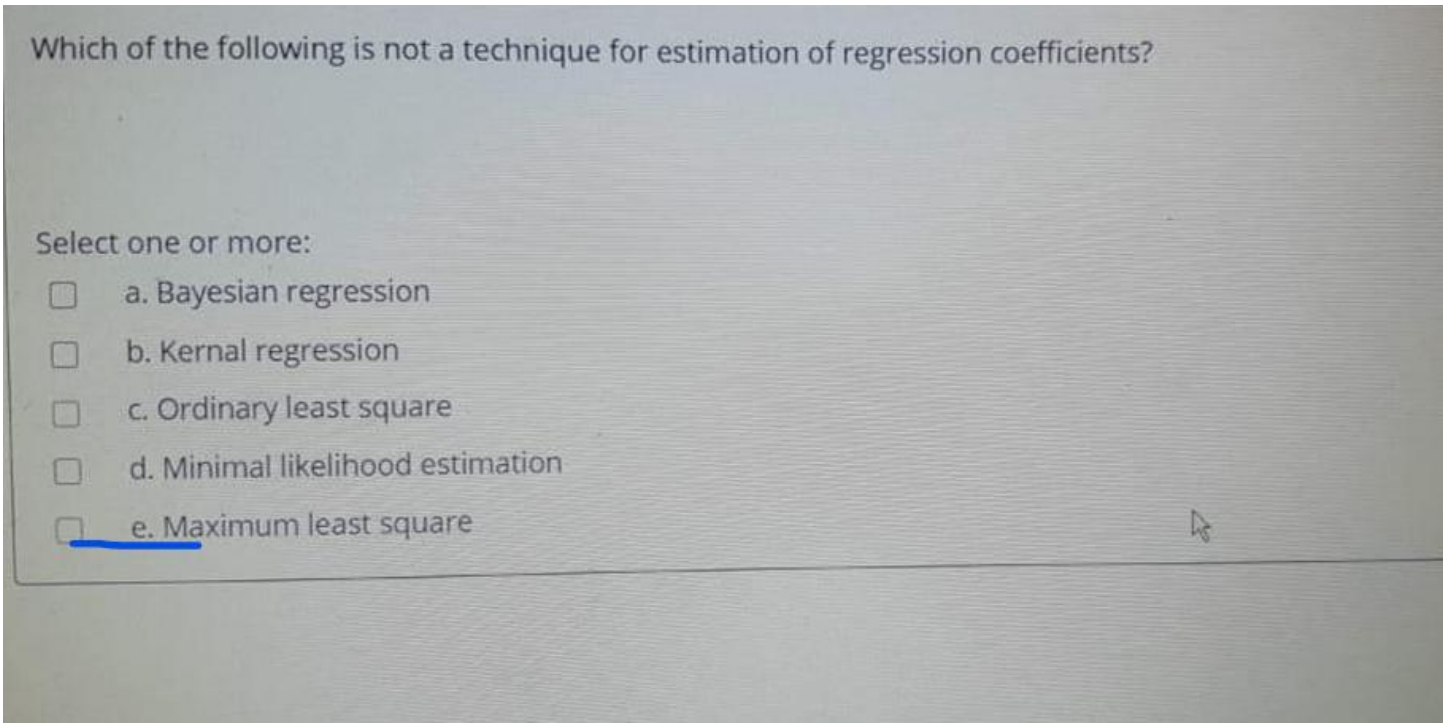

Select one:

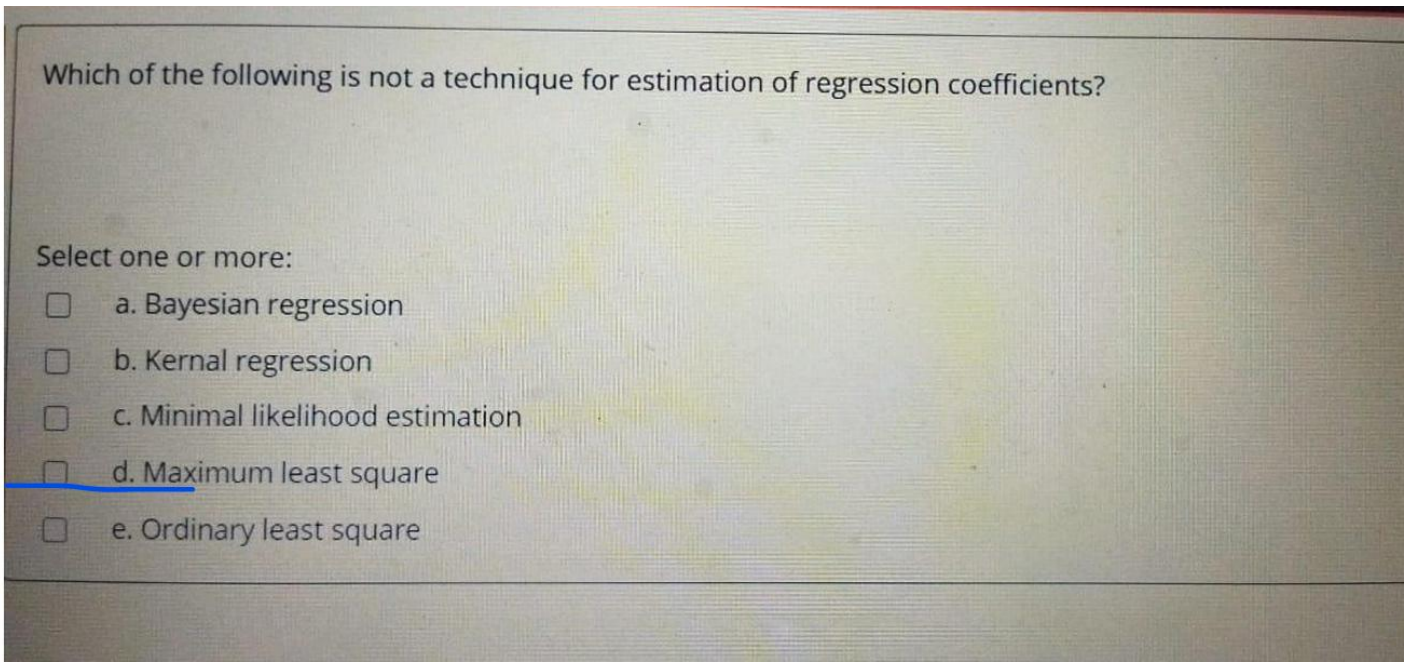
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- e. Cars of engine displacement more than 3 liters will always be considered to be bought

# $R^2$ / Coefficient of Determination



# Regression coefficient estimation



Under which conditions could simple linear regression could be applied?

Select one or more:

- [ ] a. The mean value for error is a constant
- [ ] b. The mean value for regression coefficient is a constant
- [ ] c. There should be a linear relationship between dependent and independent variable
- [ ] d. The variance of the errors is constant
- [ ] e. The independent variable and errors follow a normal distribution